

(a) 4-site: zeros (%)

a	51.2	51.8	88.5
m	52.7	52.4	60.4
h	81.3	97.9	93.3
q	0.0	0.0	0.0
k	0.0	0.0	0.0
v	0.0	0.0	0.0
z	77.6	88.3	97.4
	P_0	P_h	P_{all}

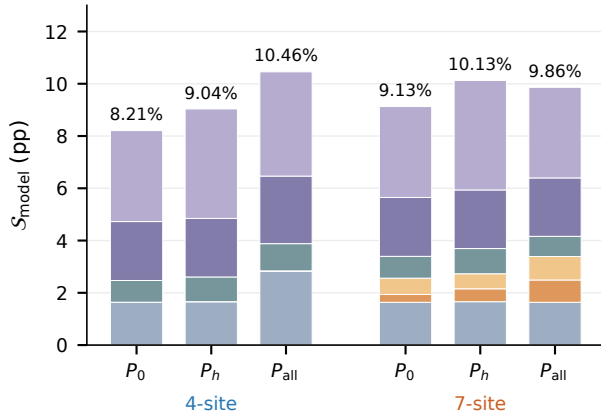
Pressure scope

(b) 7-site: zeros (%)

a	51.0	51.8	51.0
m	52.7	52.4	52.2
h	81.3	97.9	81.1
q	1.8	2.3	5.4
k	1.7	3.6	5.8
v	7.3	6.7	10.5
z	78.0	90.1	72.1
	P_0	P_h	P_{all}

Pressure scope

(c) Operation contributions



QKV QK PV Attention output FFN up FFN down